

Control work 2, and work on the mistakes

The integers in one paper — the four operations, the signs and the order.

LESSON 35–36

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 2

S8 1 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions

3 min

The paper

40 min

Answers

12 min

Diagnosis and rewrite

20 min

The map

5 min

LEARNING OBJECTIVES

- Use the sign rules for all four operations under time.
- Apply the order of operations to a mixed expression.
- Compare and order integers, and use the modulus.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National

pp. 36–94

Cambridge

Stage 8, pp. 2–10

Workbook

Control paper M2

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Put in ascending order: $-7, 3, 0, -12, 5$	4	L23–24
2	Find $ -9 $, $-(-6)$ and the opposite of 11	4	L21–22
3	Compute $(-8) + (-5)$, $12 + (-17)$, $-4 - (-9)$	6	L25–28
4	Compute $(-6)(-7)$, $(-45) \div 9$, $(-2)^4$	6	L29–32
5	Evaluate $-15 \div 3 + (-2) \times 4$	5	L33–34
6	Solve $ x = 7$ and find the mean of $-5, 1, -2, 6$	5	L21–22, 33–34

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for putting the negatives in the right order among themselves; Q4 two for the even power; Q5 two for doing $\div$ and $\times$ before $+$; Q6 two for giving **both** solutions.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
negatives ordered forwards	$-7, -12$	$-12, -7$
modulus left negative	$ -9 = -9$	9
same signs subtracted	$(-8) + (-5) = -3$	-13
sign taken from the first number	$12 + (-17) = 5$	-5
double minus missed	$-4 - (-9) = -13$	5
even power made negative	$(-2)^4 = -16$	16
left to right regardless	$-15 \div 3 + (-2) \times 4 = -28$	-13

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter II as one map

BLOCK	THE SENTENCE
why negatives exist	so that every subtraction has an answer
the coordinate line	further right means greater
the modulus	the distance from zero — never negative
addition	same signs add and keep; different signs subtract and take the larger sign
subtraction	$a - b = a + (-b)$
multiplication and division	alike gives +, unlike gives -
powers	even exponent positive, odd exponent keeps the sign
order	brackets, powers, $\times \div$, $+-$

LOOKING FORWARD

The next chapter fills the last gap in the number system for this year: the **rational** numbers, where division always works too. The sign rules carry over unchanged — only the arithmetic of fractions is added.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $(-8) + (-5)$, $12 + (-17)$, $-4 - (-9)$.

① Same signs: $8 + 5 = 13$, keep the minus — -13 .

② Different signs: $17 - 12 = 5$, sign of -17 — -5 .

③ Rewrite: $-4 + 9$.
Two minuses.

④ $= 5$

ANSWER

$-13, -5, 5$

EXAMPLE 2 Model answer, Q5: $-15 \div 3 + (-2) \times 4$.

① Division and multiplication first.

② $-15 \div 3 = -5$

③ $(-2) \times 4 = -8$

④ $-5 + (-8) = -13$

ANSWER

-13

EXAMPLE 3 Model answer, Q6: solve $|x| = 7$ and find the mean of $-5, 1, -2, 6$.

① Two points are 7 from zero.

② $x = 7$ or $x = -7$.

Both needed.

③ Sum: $-5 + 1 - 2 + 6 = 0$.

④ Mean = $0 \div 4 = 0$.

ANSWER

$x = \pm 7$; mean 0

03 Interactive model

Mark Q3 aloud, reading each sum as a sentence about money; every wrong sign becomes obviously wrong before any rule is quoted.

Chapter II in twelve questions

Negatives exist so that:

addition works

subtraction always works

division works

nothing

Zero is:

positive

negative

neither

both

Greater: -8 or -3 ?

-8

-3

equal

cannot tell

$|-9|$ equals:

-9

9

0

81

$(-8) + (-5)$ equals:

-3

-13

3

13

$12 + (-17)$ equals:

5

-5

29

-29

$a - b$ equals:

$a + b$

$a + (-b)$

$b - a$

$-a - b$

$-4 - (-9)$ equals:

-13

5

-5

13

$(-6)(-7)$ equals:

42

-42

13

-13

$(-45) \div 9$ equals:

5

-5

-405

405

$(-2)^4$ equals:

16

-16

8

-8

Order of operations:

left to right always

brackets, powers, $\times \div$, $+-$

$+-$ first

no rule

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Integer	Butun son	Целое число
Modulus	Modul	Модуль
Sign rule	Ishoralar qoidasi	Правило знаков
Order of operations	Amallar tartibi	Порядок действий
Ascending order	O'sish tartibi	По возрастанию
Opposite number	Qarama-qarshi son	Противоположное число
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests mainly:

arithmetic

ordering negatives

speed

the modulus

In Q3 the third part needs:

the addition rule

the double-minus rule

a power

a modulus

$(-2)^4$ is positive because:

the base is even

the exponent is even

the base is negative

no reason

Q5 requires:

left to right

$\times$ and $\div$ first

$+$ and $-$ first

brackets

Q6's equation has:

one answer

two answers

no answer

many

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Ascending: $-7, 3, 0, -12, 5$

ANSWER $-12, -7, 0, 3, 5$

2. $|-9|$

ANSWER 9

3. $-(-6)$

ANSWER 6

4. Opposite of 11

ANSWER -11

5. $(-8) + (-5)$

ANSWER -13

6. $12 + (-17)$

ANSWER -5

7. $-4 - (-9)$

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. $(-6)(-7)$

ANSWER 42

2. $(-45) \div 9$

ANSWER -5

3. $(-2)^4$

ANSWER 16

4. $-15 \div 3 + (-2) \times 4$

ANSWER -13

5. Solve $|x| = 7$

ANSWER ± 7

6. Mean of $-5, 1, -2, 6$

ANSWER 0

7. $(-3)^3$

ANSWER -27

Hard · stretch and reason

7 PROBLEMS

1. $(-2)^3 + (-3)^2 - (-4)$

ANSWER 5

2. $-24 \div (-6) \times (-2)$

ANSWER -8

3. $(-5)(3) - (-2)(-6)$

ANSWER -27

4. Mean of $-9, -3, 0, 6, 11$

ANSWER 1

5. Solve $|x| \leq 2$ in integers

ANSWER $-2, -1, 0, 1, 2$

6. Find x : $-4x + 3 = 19$

ANSWER -4

7. Evaluate $2a - b^2$ at $a = -5, b = -3$

ANSWER -19

07 Homework**Homework — 5 tasks**

Rewrite in full every question that lost a mark before the next chapter begins.

1. Put in ascending order: $-4, 8, 0, -15, 2$.
2. Compute $(-9) + (-6)$, $14 + (-21)$ and $-7 - (-12)$.
3. Compute $(-8)(-4)$, $(-56) \div 7$ and $(-3)^4$.
4. Evaluate $-18 \div 6 + (-3) \times 5$.
5. Solve $|x| = 12$ and find the mean of $-7, 2, -4, 9$.